

PAPER_1326_SMBH_SEEDS

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PAPER_1326 — Supermassive BH Seed Mass via Direct Collapse

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Framework: UQFF — Star-Magic v5.27+

Tier: D — Astrophysics / Cosmology Open (CLOSED)

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Calculator: `calculate_paradox({"paradox": "smbh_seeds"})`

Master Expression $M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 56,160 M_{\text{sun}}$
****Match: within obs $10^{10} - 10^{12} M_{\text{sun}}$ ****

Interpretation Direct collapse seeds via DPM buoyancy at $z > 15$.
Explains $z \sim 10$ SMBH observations.

UQFF Locked Primitives - $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$ - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.

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